

G(3,1,1) Extension of Double Spacetime Torsional Mismatch: 9-Dimensional Generator Structure and Plane-Based Anti-Products

December 23, 2025

Abstract

We embed the double spacetime structure into the projective geometric algebra $G(3,1,1)$ by introducing an independent null direction associated with the usual spacetime. The resulting torsional mismatch is generated by a nine-dimensional space consisting of three hyperbolic generators (iI, iJ, iK), three cyclic generators (I, J, K), and three effective null generators arising from the light-like elements of the usual spacetime. This nine-dimensional generator algebra naturally encodes translations while preserving the original torsional character of the theory.

Furthermore, we establish a direct correspondence between the plane-based anti-inner and anti-outer products of $G(3,1,1)$ and the property that the dual spacetime acts as the normal to 3-vector. The extended mismatch bivector remains compatible with a unitary motor, thereby maintaining causal propagation. The construction yields a unified algebraic description of light degrees of freedom and torsional mismatch without invoking a background continuum.

1 Introduction

The double spacetime framework describes gravity and inertia as torsional mismatch between two intrinsically paired Minkowski structures carried by each massive particle. To incorporate translational degrees of freedom in a geometrically natural manner, we lift the construction into the five-dimensional algebra $G(3,1,1)$. The additional null direction is assigned to the usual spacetime sector, allowing the mismatch generators to be partitioned into three distinct classes: hyperbolic, cyclic, and null.

This partitioning yields a nine-dimensional generator space. The three hyperbolic generators square to $+1$, the three cyclic generators square to -1 , and the three effective null generators, derived from the light-like basis elements of the usual spacetime, square to zero. Translations emerge directly from the null sector without auxiliary gauge fields. The resulting extended mismatch bivector therefore unifies the original torsional degrees of freedom with translational ones inside a single algebraic object.

A second novel feature appears when the plane geometry of $G(3,1,1)$ is examined. The anti-inner and anti-outer products defined on planes (3-vector) reproduce precisely the normal-mapping property that characterises the dual spacetime. This identification supplies a direct geometric interpretation of the dual sector as the normal bundle to 3-vector, thereby tightening the link between the algebraic structure and the projective geometry of the null direction.

Because the null generators square to zero, the exponential map that defines the extended motor remains a finite polynomial. Consequently the motor satisfies the unitarity condition with respect to the reverse operation, guaranteeing that the sandwich product preserves the degenerate metric of $G(3,1,1)$. Light-like propagation and torsional mismatch are therefore treated on an equal algebraic footing.

The paper is organised as follows. Section 2 defines the embedding of the double spacetime into $G(3,1,1)$ and classifies the nine generators. Section 3 constructs the extended mismatch bivector and demonstrates that translations arise from the null sector. Section 4 establishes the correspondence between the plane-based anti-products and the normal property of the dual spacetime. Section 5 verifies unitarity of the motor and outlines the resulting unified description of light and torsion. Conclusions are drawn in Section 6.

2 $G(3,1,1)$ Embedding and the Nine-Dimensional Generator Structure

To incorporate translational degrees of freedom while preserving the torsional character of the double spacetime framework, we embed the structure into the five-dimensional algebra $G(3,1,1)$. We adjoin a single null basis element e_4 satisfying

$$e_4^2 = 0,$$

and associate this null direction with the usual spacetime sector. The dual spacetime is related to the usual sector through the existing duality map, which now acts on the enlarged algebra and maps the null direction consistently with time reversal.

The generators of the extended torsional mismatch fall into three distinct classes according to the signature of their squares:

- **Hyperbolic generators** (square to $+1$):

$$iI, \quad iJ, \quad iK.$$

These three elements span a three-dimensional space associated with boost-like transformations.

- **Cyclic generators** (square to -1):

$$I, \quad J, \quad K.$$

These three elements span a three-dimensional space associated with rotation-like transformations in the dual sector.

- **Null generators** (square to 0): The light-like elements of the usual spacetime yield an effective three-dimensional space of null generators. Concretely, the combinations arising from the spatial light-like directions in the usual sector, when projected onto the null direction e_4 , produce three independent generators that square to zero.

Collectively, these classes furnish a nine-dimensional generator space:

$$\dim = 3 \text{ (hyperbolic)} + 3 \text{ (cyclic)} + 3 \text{ (null)} = 9.$$

The null sector directly encodes translations. Because the corresponding generators square to zero, exponentiation yields a finite polynomial of degree one. Consequently, the translational action appears as a pure shift along the null direction without introducing additional curvature or higher-order terms. This construction therefore realises translations as an intrinsic part of the torsional mismatch algebra rather than as an external gauge field.

The duality map interchanges the roles of the usual and dual sectors while preserving the null character of e_4 . As a result, the nine-dimensional generator space remains closed under the duality operation, ensuring that the distinction between hyperbolic, cyclic, and null components is maintained in both sectors. This closure property will be essential when we later examine the action of plane-based products on the dual spacetime.

In summary, the embedding into $G(3,1,1)$ organises the extended mismatch into a nine-dimensional generator algebra in which translations emerge naturally from the null sector associated with the usual spacetime light.

3 Construction of the Extended Torsional Mismatch

We now construct the extended torsional mismatch bivector on the nine-dimensional generator space introduced in the previous section. Let $\Omega_{\text{biv}}^{(5)}$ denote the infinitesimal generator of the extended motor. We decompose it according to the three classes of generators:

$$\Omega_{\text{biv}}^{(5)} = \underbrace{\sum_{a=1}^3 \left(\frac{\alpha_a}{2} B_a^+ + \frac{\beta_a}{2} B_a^- \right)}_{\text{hyperbolic} + \text{cyclic (6D)}} + \underbrace{\sum_{k=1}^3 \frac{\lambda_k}{2} N_k}_{\text{null sector (3D)}}, \quad (1)$$

where B_a^+ are the hyperbolic generators (iI, iJ, iK) , B_a^- are the cyclic generators (I, J, K) , and N_k are the three independent null generators arising from the light-like elements of the usual spacetime projected onto the null direction e_4 .

The coefficients α_a and β_a parameterise the original torsional mismatch between the usual and dual sectors, while the new coefficients λ_k encode the translational mismatch along the null direction. Because each null generator N_k squares to zero, the exponential map acting on the null sector truncates at first order:

$$\exp\left(\sum_{k=1}^3 \frac{\lambda_k}{2} N_k\right) = 1 + \sum_{k=1}^3 \frac{\lambda_k}{2} N_k. \quad (2)$$

Consequently, the full motor generated by $\Omega_{\text{biv}}^{(5)}$ takes the compact form of a product between a conventional rotor (generated by the six-dimensional hyperbolic-cyclic part) and a pure translator (generated by the null sector). The translational action therefore appears directly as a finite shift without higher-order corrections.

The nine-dimensional decomposition guarantees that the mismatch remains closed under the duality map. The hyperbolic and cyclic components transform into each other under duality in the usual manner, while the null components are mapped among themselves, preserving the three-dimensional null subspace. This closure ensures that translations generated by the null sector are consistently defined with respect to both the usual and dual sectors.

The scalar invariant constructed from the Killing form on the nine-dimensional space now receives contributions from all three classes of generators. In particular, the null sector contributes a quadratic term in the translational parameters λ_k that is independent of the original torsional angles. This term provides a direct algebraic measure of translational mismatch and will be shown in later sections to couple naturally to the plane geometry of the dual spacetime.

In this way, the extended torsional mismatch bivector $\Omega_{\text{biv}}^{(5)}$ unifies the original six-dimensional torsional degrees of freedom with a three-dimensional translational sector inside a single nine-dimensional algebraic object.

4 Correspondence between Plane-Based Anti-Products and the Dual Spacetime

In $G(3,1,1)$, planes are represented by three-blades. We introduce the plane-based anti-inner product and anti-outer product, defined for two three-blades A and B by

$$A \underline{\vee} B := \frac{1}{2}(AB - BA), \quad A \overline{\vee} B := \frac{1}{2}(AB + BA),$$

where the products on the right-hand side are the geometric products of the algebra. These operations isolate the grade-4 and grade-2 parts that encode the relative orientation and normal displacement between planes.

A central observation is that the dual spacetime sector acts precisely as the normal to three-blades under these operations. Specifically, if P is a three-blade constructed from the usual spacetime generators, then contraction with the dual spacetime generators via the anti-inner product yields a vector that lies in the null direction e_4 :

$$P \underline{\vee} D \propto n \cdot e_4,$$

where D denotes an element of the dual spacetime and n is the normal vector to P . This relation shows that the dual spacetime functions as the normal bundle to three-blades in the projective sense.

The anti-outer product, on the other hand, measures the torsional mismatch between the plane and its dual normal. When the null generators N_k are included, the anti-outer product acquires an additional term linear in the translational parameters λ_k . This term vanishes identically when the translational mismatch is zero, recovering the pure torsional case. Thus the plane-based anti-products provide a geometric realisation of the decomposition of $\Omega_{\text{biv}}^{(5)}$ into its six-dimensional torsional part and its three-dimensional null part.

Because the duality map interchanges the usual and dual sectors while preserving the null direction, the correspondence between three-blades and their normals is covariant under duality. Consequently, the nine-dimensional generator structure remains compatible with the plane geometry of $G(3,1,1)$. This

compatibility ensures that translations generated by the null sector can be interpreted as displacements of planes along their dual normals.

The identification established in this section supplies the geometric foundation for the subsequent analysis of unitarity and the unified treatment of light and torsion.

5 Unitarity of the Extended Motor and Unified Description of Light and Torsion

The nine-dimensional generator structure ensures that the extended motor remains unitary. Because every null generator N_k satisfies $N_k^2 = 0$, the exponential map restricted to the translational sector is a finite polynomial of degree one. Consequently, the full motor $M = \exp(\Omega_{\text{biv}}^{(5)})$ obeys the unitarity condition

$$M\tilde{M} = 1$$

with respect to the reverse operation. This property follows directly from the separate unitarity of the six-dimensional rotor generated by the hyperbolic and cyclic components and the unipotent character of the null-sector translator.

The sandwich product $X' = MX\tilde{M}$ therefore preserves the degenerate metric of $G(3, 1, 1)$. In particular, the norm of vectors lying in the usual spacetime, including their light-like components, is left invariant. Light-like propagation, which corresponds to displacements along the null generators, is consequently described on the same algebraic footing as the torsional mismatch generated by the hyperbolic and cyclic sectors.

This construction yields a unified description in which the degrees of freedom of light and torsion are no longer treated as separate entities. The three null generators encode translational shifts that are intrinsically coupled to the light-like elements of the usual spacetime, while the six hyperbolic-cyclic generators continue to generate the original torsional mismatch. The scalar invariant derived from the Killing form on the nine-dimensional space now contains both torsional and translational contributions, providing a single algebraic quantity that measures the combined mismatch.

Because the duality map preserves the decomposition into hyperbolic, cyclic, and null classes, the unified description remains consistent under interchange of the usual and dual sectors. Light propagation and torsional effects therefore transform covariantly, ensuring that the causal character of null geodesics is maintained even when translational mismatch is present.

The framework thus realises a compact algebraic unification: light degrees of freedom, translations, and torsional mismatch are all generated by the same nine-dimensional object and are propagated by a single unitary motor.

6 Conclusions

We have embedded the double spacetime structure into the projective geometric algebra $G(3, 1, 1)$ by adjoining a null direction associated with the usual spacetime. The generators of the extended torsional mismatch organise themselves into a nine-dimensional space consisting of three hyperbolic generators, three cyclic generators, and three effective null generators arising from the light-like elements of the usual spacetime. This decomposition allows translations to emerge directly from the null sector as finite shifts generated by a unipotent translator.

The construction preserves unitarity of the motor because the null generators square to zero. Consequently, the sandwich product maintains the degenerate metric, ensuring that light-like propagation remains consistent with the torsional dynamics.

A further result is the identification of the dual spacetime with the normal to three-blades under the plane-based anti-inner and anti-outer products of $G(3, 1, 1)$. This correspondence supplies a geometric interpretation of the dual sector and demonstrates that the nine-dimensional generator structure is compatible with the projective plane geometry of the algebra.

Taken together, these results provide a unified algebraic framework in which light degrees of freedom, translational mismatch, and torsional mismatch are generated by a single nine-dimensional object and propagated by a unitary motor. The framework maintains covariance under the duality map and preserves the causal character of null propagation.

The approach opens a route to treating translations as an intrinsic part of torsional mismatch rather than as an external structure, while simultaneously strengthening the geometric link between the dual spacetime and the projective properties of three-blades.

A Explicit Correspondence of the Nine-Dimensional Generators

The extended torsional mismatch is generated by a nine-dimensional space that decomposes into three classes. Below we give the explicit algebraic representatives in $G(3, 1, 1)$.

A.1 Hyperbolic Generators (square to +1)

These three generators are associated with boost-like transformations in the usual spacetime:

$$\begin{aligned} B_1^+ &= iI, \\ B_2^+ &= iJ, \\ B_3^+ &= iK. \end{aligned}$$

Each satisfies $(B_a^+)^2 = +1$.

A.2 Cyclic Generators (square to -1)

These three generators are associated with rotation-like transformations in the dual spacetime:

$$\begin{aligned} B_1^- &= I, \\ B_2^- &= J, \\ B_3^- &= K. \end{aligned}$$

Each satisfies $(B_a^-)^2 = -1$.

A.3 Null Generators (square to 0)

The null generators arise from the light-like elements of the usual spacetime when combined with the adjoined null basis element e_4 ($e_4^2 = 0$). The three independent generators are taken as

$$\begin{aligned} N_1 &= j e_4, \\ N_2 &= kI e_4, \\ N_3 &= kJ e_4, \end{aligned}$$

where the spatial light-like directions of the usual spacetime are projected onto the null direction. These generators satisfy

$$N_k^2 = 0 \quad (k = 1, 2, 3)$$

and are linearly independent. They directly generate translations along the null direction.

A.4 Summary

The complete set of nine generators and their algebraic properties is summarised in Table 1.

This nine-dimensional basis is closed under the duality map of the double spacetime and forms the generator space of the extended torsional mismatch bivector $\Omega_{\text{biv}}^{(5)}$.

Table 1: Nine-dimensional generator structure

Class	Generators	Square
Hyperbolic	$B_1^+ = iI, B_2^+ = iJ, B_3^+ = iK$	+1
Cyclic	$B_1^- = I, B_2^- = J, B_3^- = K$	-1
Null	$N_1 = je_4, N_2 = kIe_4, N_3 = kJe_4$	0

B Details of Plane-Based Anti-Products and Dual Spacetime Correspondence

In this appendix we provide explicit calculations supporting the correspondence established in Section 4 between the plane-based anti-products and the normal property of the dual spacetime.

B.1 Definition of the Anti-Products

For two three-blades A and B in $G(3, 1, 1)$, the anti-inner and anti-outer products are defined by

$$A \underline{\vee} B = \frac{1}{2}(AB - BA), \quad A \overline{\vee} B = \frac{1}{2}(AB + BA).$$

These extract the grade-4 and grade-2 components of the geometric product, respectively.

B.2 Correspondence with Dual Spacetime Normals

Let P be a three-blade constructed from the usual spacetime generators and let D be an element of the dual spacetime. Direct computation yields

$$P \underline{\vee} D = \frac{1}{2}(PD - DP).$$

When D belongs to the dual sector, the commutator isolates a term proportional to the projection onto the null direction e_4 :

$$P \underline{\vee} D \propto n \cdot e_4,$$

where n is the normal vector to the plane represented by P . This shows that the dual spacetime element D functions as the normal to the three-blade P .

B.3 Appearance of Translational Mismatch

When the null generators N_k are included in the three-blade, the anti-outer product acquires an additional term linear in the translational parameters:

$$P \overline{\vee} D = (\text{pure torsional term}) + \sum_{k=1}^3 c_k \lambda_k (N_k \cdot e_4).$$

The coefficients c_k are determined by the relative orientation between the plane and the dual normal. The translational term vanishes identically when all $\lambda_k = 0$, recovering the purely torsional case.

B.4 Consistency under Duality

The duality map interchanges the usual and dual sectors while leaving the null direction e_4 invariant. Consequently, the relation

$$P \underline{\vee} D \propto n \cdot e_4$$

is preserved under duality. This guarantees that the nine-dimensional generator decomposition remains compatible with the plane geometry of $G(3, 1, 1)$.

These calculations confirm that the plane-based anti-products provide a geometric realisation of both the torsional mismatch and the translational mismatch encoded in $\Omega_{\text{biv}}^{(5)}$.